

Spectral Attention-Enhanced HybridsN with Multi-Objective Loss for Underwater Habitat Segmentation from Hyperspectral Imagery

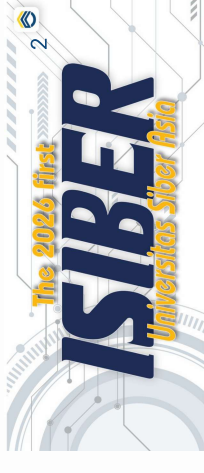
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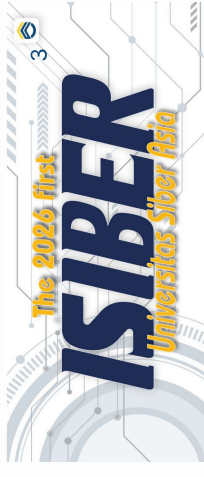
26 February 2026

Outlines

1. **Background** (Motivations, Problems, Research Objectives, Related Works, Gaps)
2. **Methodology** (Dataset Overview, Preprocessing Pipeline, Baseline Architecture, Proposed Novelties,
Training Strategy)
3. **Results** (Ablation Study Results, Spectral Attention Analysis, Failure Analysis, Qualitative Results, Key Findings,
Contributions, Limitations, Future Works)

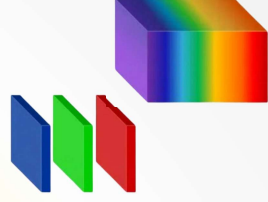
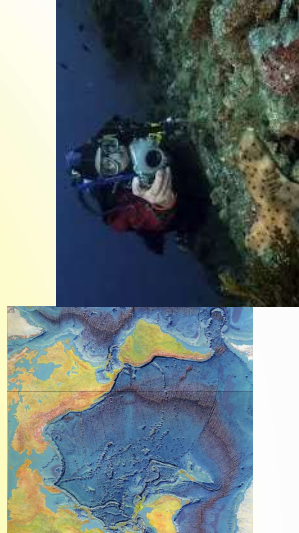


Motivation & Problem



The Challenges

- < 10% ocean floors mapped
- Manual surveys are expensive and slow
- Insufficient spectral information when using RGB Image



Why it matters?



- Climate change monitoring
- Marine biodiversity conservation
- Sustainable resource management



Technical Problem

- Severe class imbalance (74% background)
- Similar spectral signatures
- Water absorption dominates IR bands



Research Objectives

1

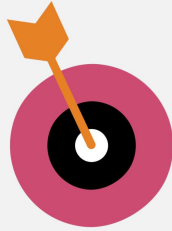
Enhance HybridSN with Spectral Self-Attention

→ Adaptively emphasize underwater-relevant wavelengths

2

Investigate Multi-Objective Loss Combinations

→ Address extreme class imbalance



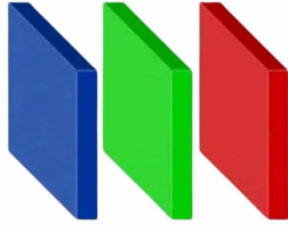
3

Validate on Real Norwegian Coastal Dataset

→ 7 habitat classes, 300 spectral bands



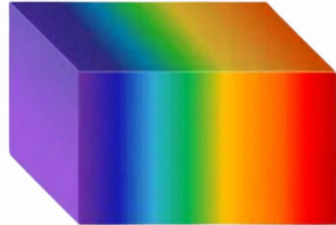
Hyperspectral Imaging for Underwater



RGB Camera :

3 Bands (Red, Green, Blue)

→ Limited discrimination



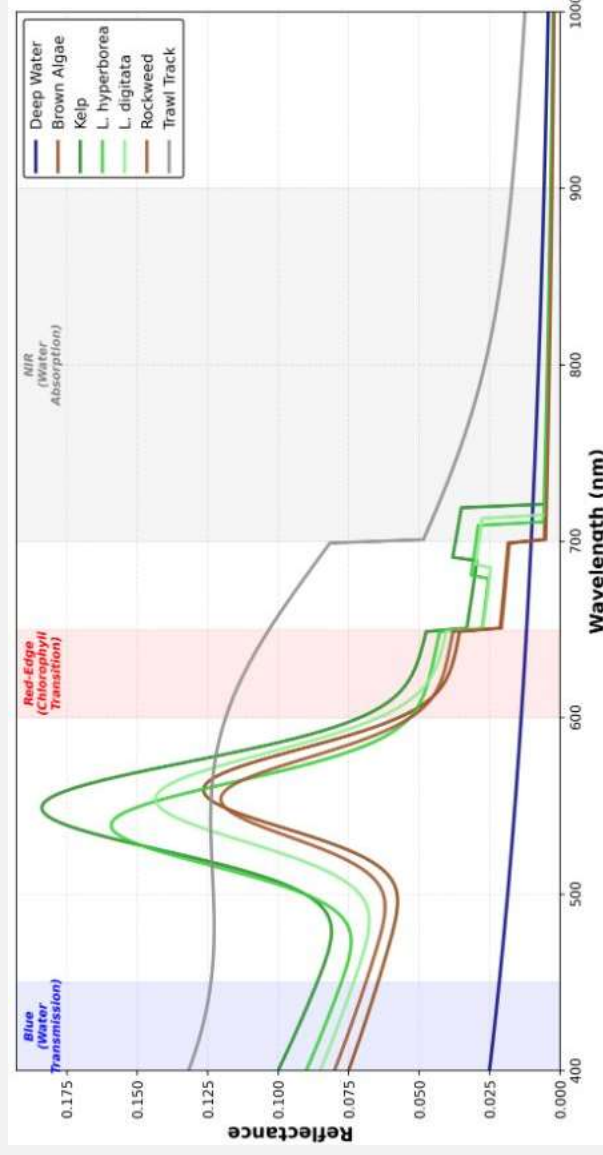
Hyperspectral :

300 Bands
[400nm...500nm...1000nm]

→ Rich spectral features

Dataset for this study :

- Sensor : Resonon Pika-L
- Bands: 300 (400-1000 nm)
- GSD: 3.5 cm
- Platform: UAV-mounted
- Location: Norwegian coast



Related Works & Gaps

Classical ML (2010-2015)

- SVM, Random Forest
 - Fast
 - No spatial context

- Handcrafted features
 - Domain expertise required
 - Interpretable

Deep Learning (2016-2023)

- 2D CNN
 - Spatial features
 - Ignores spectral

- 3D CNN
 - Spectral-spatial
 - High computation
 - Easy overfitting

- HybridSN
 - Efficient 3D+2D
 - Uniform band weight
 - Terrestrial focus

Our Work (2025)

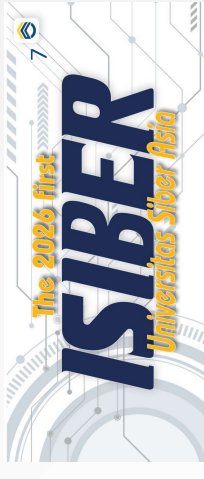
- HybridSN + Spectral Attention
 - Adaptive wavelength weighting
 - Negative results on loss combinatons

GAP

Existing methods : Treat all wavelengths equally

Underwater Specific : Need to suppress water-absorbed bands

Extreme Imbalance : Standard losses fail



Dataset Overview

MASSIMAL Dataset (Norwegian Coastal Sites)

- Maholmen, Smola Island, Norway
- 17 Hyperspectral images
- 300 Bands
- 3.5 cm ground sampling distance
- 7 Habitat classes
 - Deep water (background) 74%
 - Brown Algae
 - Kelp
 - Laminaria hyperborea
 - Laminaria digitata ← Rare class (1 image)
 - Rockweed ← Rare class (1 image)
 - Trawl Track

Split strategy

- Train: 8 images → 2,805 tiles (96×96)
- Val: 6 images → 3,490 tiles
- Test: 2 images → 2,338 tiles
- 1 image split vertically (contains two rare classes) and distributed for each train-val-test split

Preprocessing Pipeline

Input : 4 files per image

- radiance.bip
- radiance.bip.hdr
- irradiance.spec
- irradiance.spec.hdr

```
rad_data = base_path / f"{tile_id}_radiance.bip"
rad_hdr = rad_data.with_suffix(".bip.hdr")
irr_file = base_path / f"{tile_id}_irradiance.spec"
irr_hdr = irr_file.with_suffix(".spec.hdr")

# Load radiance cube (Langsung penuh ke RAM)
cube_obj = envi.open(str(rad_hdr), str(rad_data))
cube_radiance = cube_obj.load().astype(np.float32)
rad_wavelengths = np.array([float(w) for w in cube_obj.metadata['wavelength']])

# Load irradiance
irr = io.envi.open(str(irr_hdr), str(irr_file))
irradiance = irr.load().squeeze().astype(np.float32)
irr_wavelengths = np.array([float(w) for w in irr.metadata['wavelength']])

# Sinkronisasi irradiance
irr_selected = np.array([
    irradiance[np.abs(irr_wavelengths - wl).argmin()] for wl in rad_wavelengths
], dtype=np.float32)

# Radiance -> Reflectance
cube_reflectance = safe_divide(np.pi * cube_radiance,
                               irr_selected[np.newaxis, np.newaxis, :])

# Sun-glint removal
cube_reflectance_corrected = hedley_glint_removal(cube_reflectance,
                                                  rad_wavelengths)

# Clamp final (pasca sun-glint removal)
cube_reflectance_corrected = clamp_reflectance(cube_reflectance_corrected)
```

STEP 1 : Convert radiance to reflectance

$$\text{Reflectance} = (\pi \times \text{Radiance}) / \text{Irradiance}$$

STEP 2 : Sun glint removal

Uses NIR bands
(800-900 nm) to estimate
glint

STEP 3 : Normalization

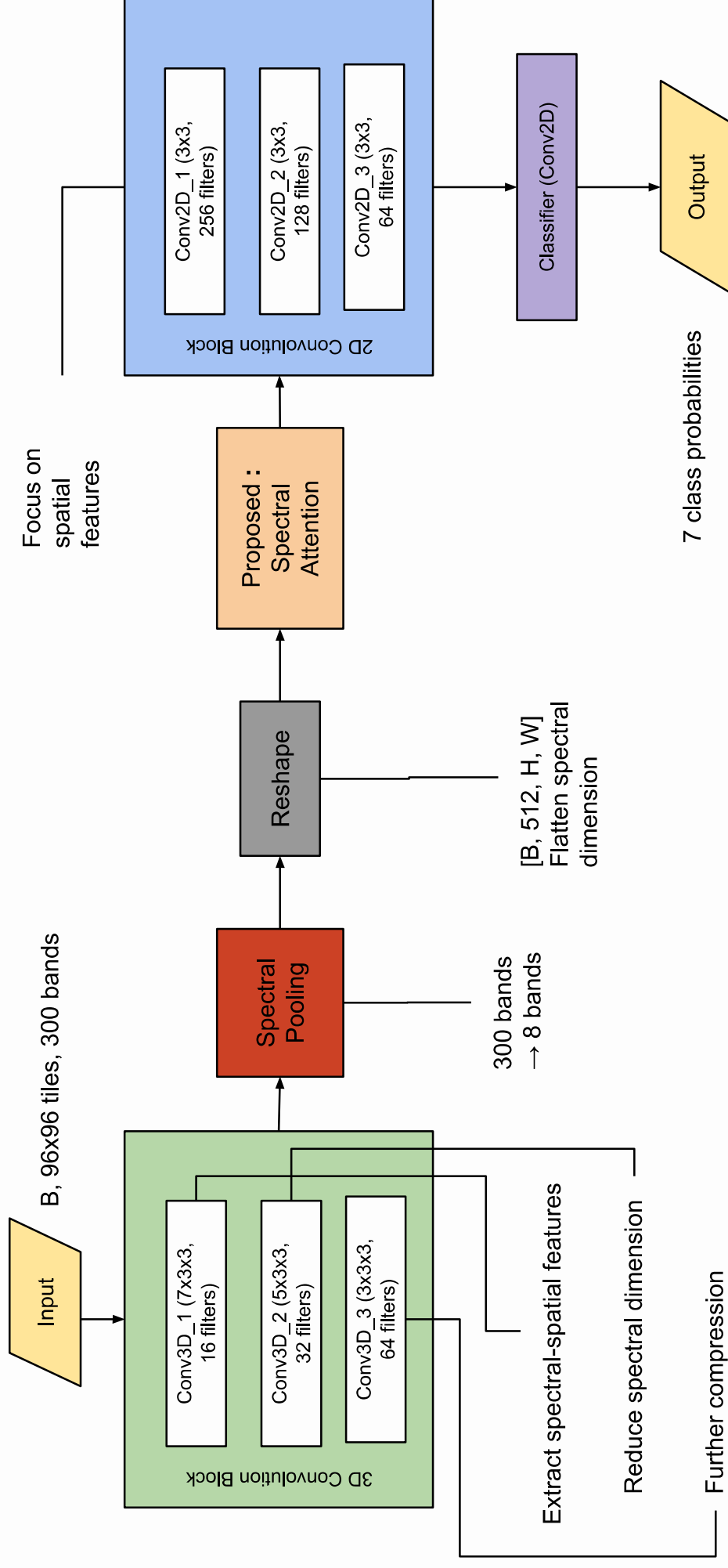
Min-Max scaling to
[0, 1]

STEP 4 : Tiling, Splitting, &
Augmentation

96×96 tiles,
augmentation : flips +
90°/180°/270°
rotations

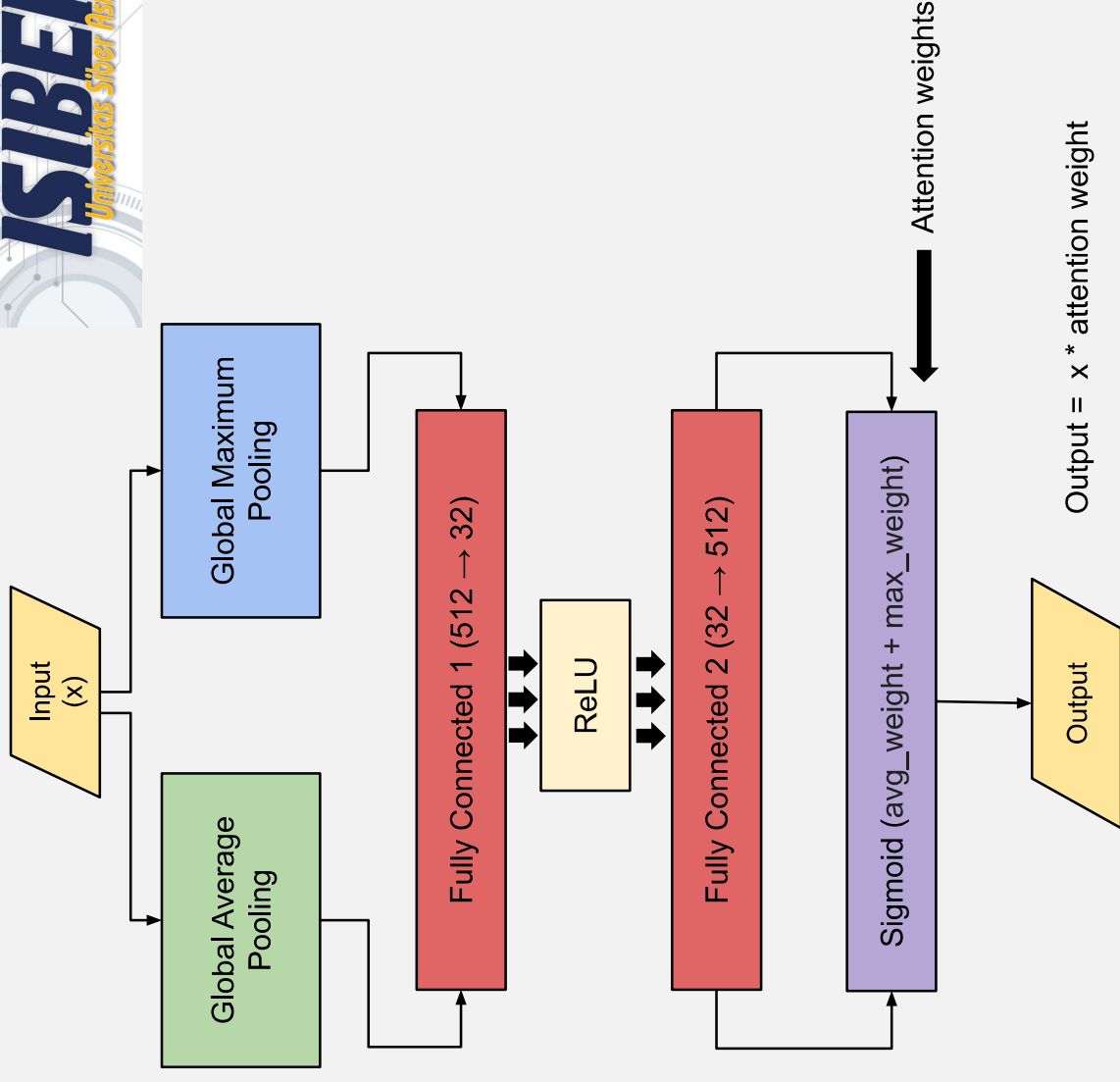
Output : Preprocessed tiles for training

Baseline Architecture (HybridSN)



Proposed Enhancement 1: Spectral Attention

- Learn which spectral combinations are discriminative
- Suppress water-absorbed bands (IR region >700nm)
- Emphasize informative regions (blue 400-450nm, red-edge 600-650nm)
- Physically interpretable attention weights



Proposed Enhancement 2 : Multi-Objective Loss



Theoretical Motivation

Focal Loss ($\gamma=5.0$)

$$L_{\text{focal}} = -\frac{1}{N} \sum_i (1 - p_{t_i})^\gamma \alpha_{t_i} \log(p_{t_i})$$

Purpose :

- Down-weight easy examples
- Focus on hard negatives
- Address class imbalance

Combined :

$$L_{\text{total}} = L_{\text{focal}} + L_{\text{dice}}$$

Dice Loss

$$L_{\text{dice}} = 1 - \frac{2 \sum_c \sum_i p_{ic} g_{ic} + \epsilon}{\sum_c \sum_i (p_{ic} + g_{ic}) + \epsilon}$$

Purpose :

- Region-level overlap
- Spatial coherence
- IoU-like optimization

Catastrophic Failure

What happened :

Baseline (CE only) : mIoU = 11.84%

- + Spectral Attention : mIoU = 12.08% $\checkmark$ (+2.0%)
- + Focal Loss ($\gamma=5.0$) : mIoU = 3.21% $\times$ (-73.4%)
- + Dice Loss : mIoU = 2.79% $\times$ (-76.9%)

Why it Failed :

- $\gamma=5.0$ over-amplifies hard examples causing gradient instability
- Focal (sample-wise) conflicts with Dice (region-wise)
- Optimizer cannot navigate chaotic loss landscape
- Attention learning disrupted

Training Strategy

Optimizer & Hyperparameters

- Optimizer : AdamW
- Learning rate : 1×10^{-4}
- Weight decay : 1×10^{-5}
- Effective batch size : 4 (via gradient accumulation)
- Gradient clipping : max norm 0.5

Learning Rate Schedule

- Phase 1 : Warmup (5 epochs)
 - Linear increase : $1 \times 10^{-5} \rightarrow 1 \times 10^{-4}$
 - Prevents early instability
- Phase 2 : ReduceLROnPlateau
 - Monitor : Validation mIoU
 - Patience : 5 epochs
 - Factor : 0.5 (halve LR)
 - Min LR : 1×10^{-7}

Early Stopping

- Patience : 15 epochs (no improvement in val mIoU)

Class Imbalance Mitigation

- Strategy 1 : Power-law class weighting
 - $\alpha_c = (1/(\text{freq}_c + \epsilon))^{0.8}$
 - Background weight = 0
- Strategy 2 : Adaptive oversampling
 - Extremely rare (<50 tiles): 15x oversample
 - Rare (50-200 tiles): 10x oversample
 - Medium (200-1000): undersample to 500-600

Data Augmentation

- Horizontal flip
- Vertical flip
- Rotations: 90°, 180°, 270°

Hardware

- Nvidia RTX 4080 Super (16GB)

Ablation Study Results

Configuration	Pixel Accuracy	mIoU	Spectral Attention	Focal Loss	Dice Loss
Baseline (CE)	56.68%	11.84%	-	-	-
+ Spectral Attention (withSA)	58.02% (+1.34%)	12.08% (+2.0%)	✓	-	-
+ Focal Loss (SAFocal)	17.64% (-69.6%)	3.21% (-73.4%)	✓	✓	-
+ Dice Loss (Full)	15.05% (-74.0%)	2.79% (-76.9%)	✓	✓	✓

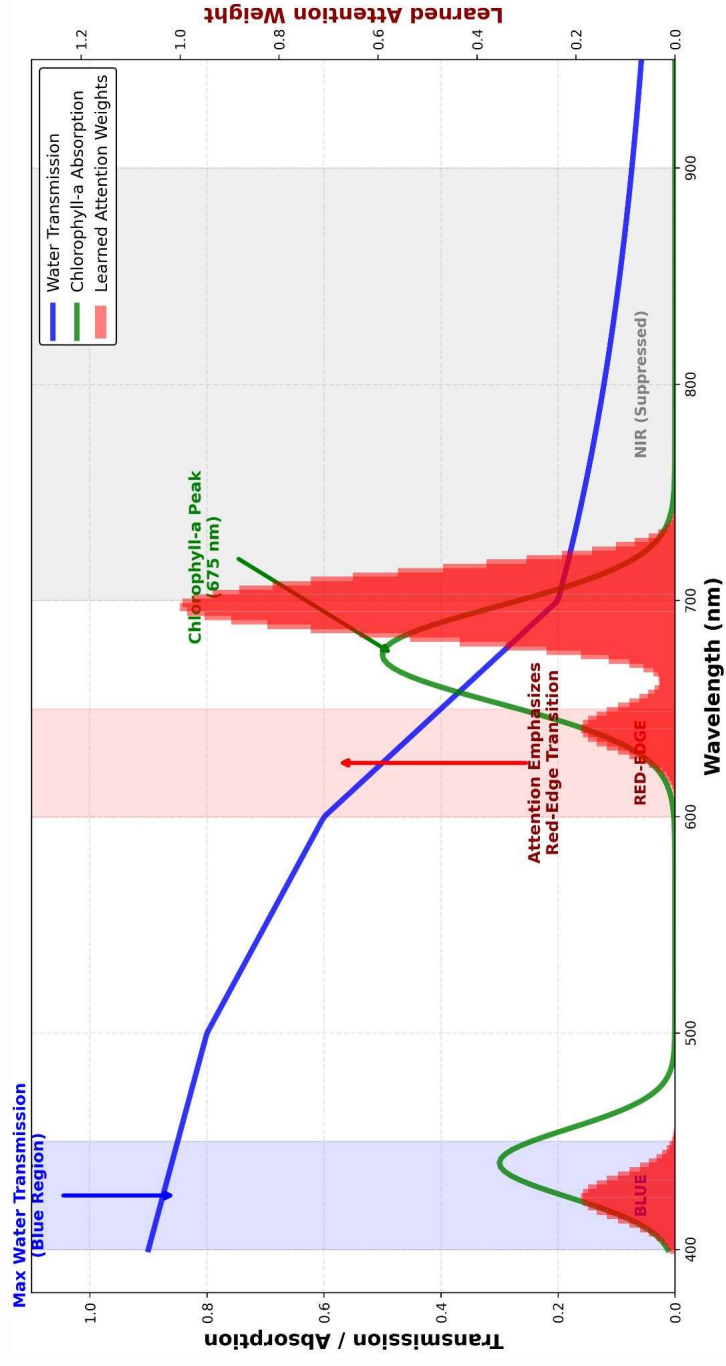
Key Findings :

- Spectral Attention : +2.0% mIoU improvement
- Stable channel selection (weights converged)
- Focal+Dice : Catastrophic optimization failure
- Model collapsed to predicting only dominant class

Spectral Attention Analysis

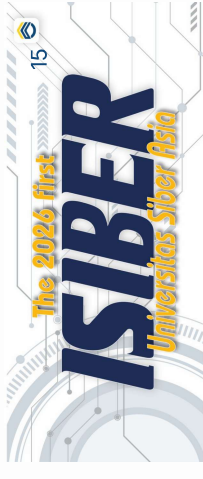
Top 9 Channels (weight = 1.0) : 152, 144, 153, 154, 147, 146, 120, 12, 448

→ Blue (400-450nm) + Red-Edge (600-650nm) Regions



- Attention aligns with underwater optical physics
- Emphasizes discriminative wavelength regions
- Network learns physically meaningful features!

Failure Analysis



Gradient Conflict

- Focal loss :
 - Amplified hard examples exponentially
 - Combined with $\alpha_c = (1/(\text{freq}_c + \epsilon))^{0.8}$ caused extremely unstable updates
- Dice loss :
 - Optimized global region overlap
 - Normalized across all pixels
- Conflict : Exponential sample-wise (Focal) met smooth region-wise (Dice) resulting chaotic loss landscape

Attention Disruption

- withSA (CE only) : Stable, consistent patterns
- SAFocal : Erratic, random noise pattern

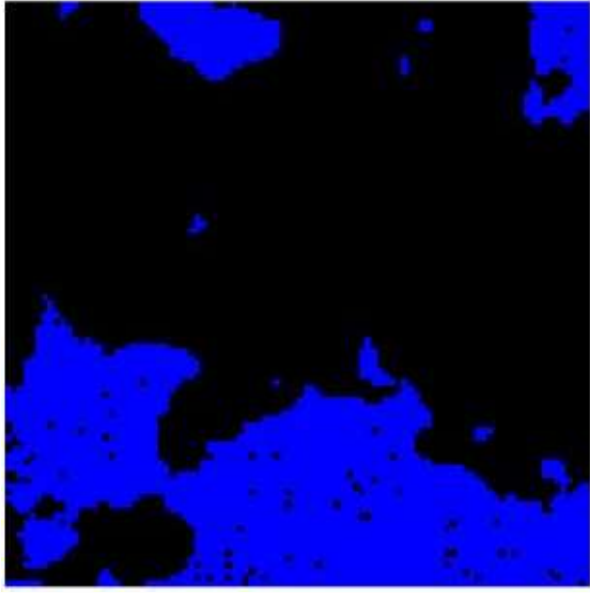
Class Collapse

- Baseline: Distributes predictions across classes
- withSA: Improved diagonal, some classes detected
- SAFocal: Collapses to Class 3 only
- Full: Complete collapse to Class 2

Lessons Learned :

- $\gamma=5.0$ too extreme for 74% imbalance
- Power-smoothed weights ($\text{freq}^{-0.8}$) amplify instability
- Multi-objective loss needs careful gradient balancing
- Simpler alternatives (class-balanced CE) may be better

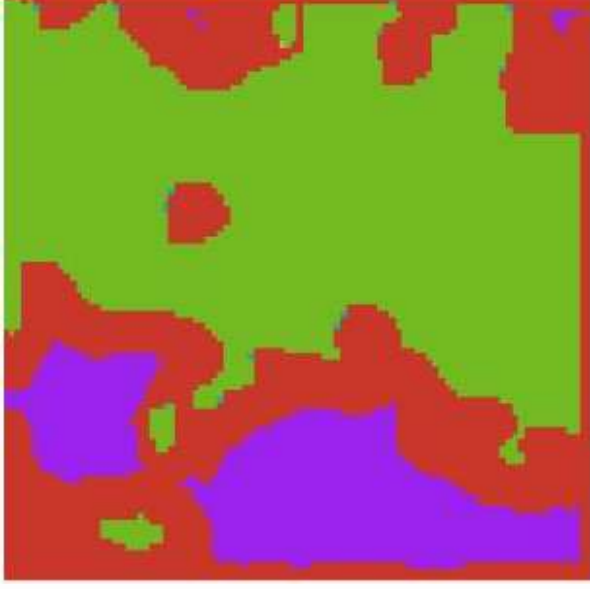
Qualitative Results



Underwater Scene

Key Observations :

- Successfully captures spatial habitat structure
- Delineates distinct ecological regions
- Preserves boundary continuity
- Demonstrates spectral-spatial feature learning



Segmentation with spatial structure

Note :

Despite class detection limitations (74% background imbalance), the model learns meaningful spatial patterns from hyperspectral features.

Key Findings & Contributions

Spectral Attention Enhancement

- +2.0% mIoU improvement (11.84% → 12.08%)
- Physically interpretable (aligns with underwater optics)
- Stable convergence across runs

Negative Result on Multi-Objective Loss

- Demonstrates fundamental incompatibility :
 - Focal ($\gamma=5.0$) + Dice → 73-77% mIoU drop
 - Gradient conflict causes optimization collapse
 - Disrupts attention learning
- Provides guidance for community :
 - Avoid extreme focusing parameters with extreme imbalance
 - Test simpler alternatives first

Underwater HSI Benchmark

- First application of HybridSN to underwater segmentation
- Dealing with 7 classes with severe imbalance (74% background)
- Real-world Norwegian coastal dataset (300 bands)

Scientific Value :

- Transparent reporting of negative results
- Physically grounded methodology
- Actionable insights for imbalanced HSI segmentation

Limitations & Future Work

Limitations

- Dataset
 - Single site (Norwegian coast only)
 - Limited size (17 images)
 - Two rare classes in single image
- Architecture
 - 96×96 tiles provide limited spatial context
 - Tile-based processing → boundary artifacts
 - Memory constraints (16GB GPU)
- Performance
 - Absolute mIoU still low (12.08%)
 - Rare class detection incomplete
 - Multi-objective loss failed

Future Directions

- Cross-Site Validation
 - Test on different geographic locations
 - Different sensor types (cross-sensor)
 - Seasonal variations
- Architecture Improvements
 - Multi-scale processing (64/96/128 tiles)
 - Hierarchical attention (band + spatial)
 - Transformer-based encoder
 - Memory-efficient training techniques
- Loss Function Research
 - Class-balanced Cross-Entropy
 - Lovász-Softmax loss
 - Sequential training strategies
 - Gradient balancing mechanisms
- Practical Applications
 - Real-time monitoring systems
 - Integration with bathymetry data
 - Decision support for conservation

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 SINERGI Hilirisasi Inovasi Komersial Program

 Ethereum Foundation Academic Grants Round 2025

 Dataset: MASSIMAL (Skjelvareid et al., 2025) NIRD Research Data Archive

Thank You For Your Attention, any questions?

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